

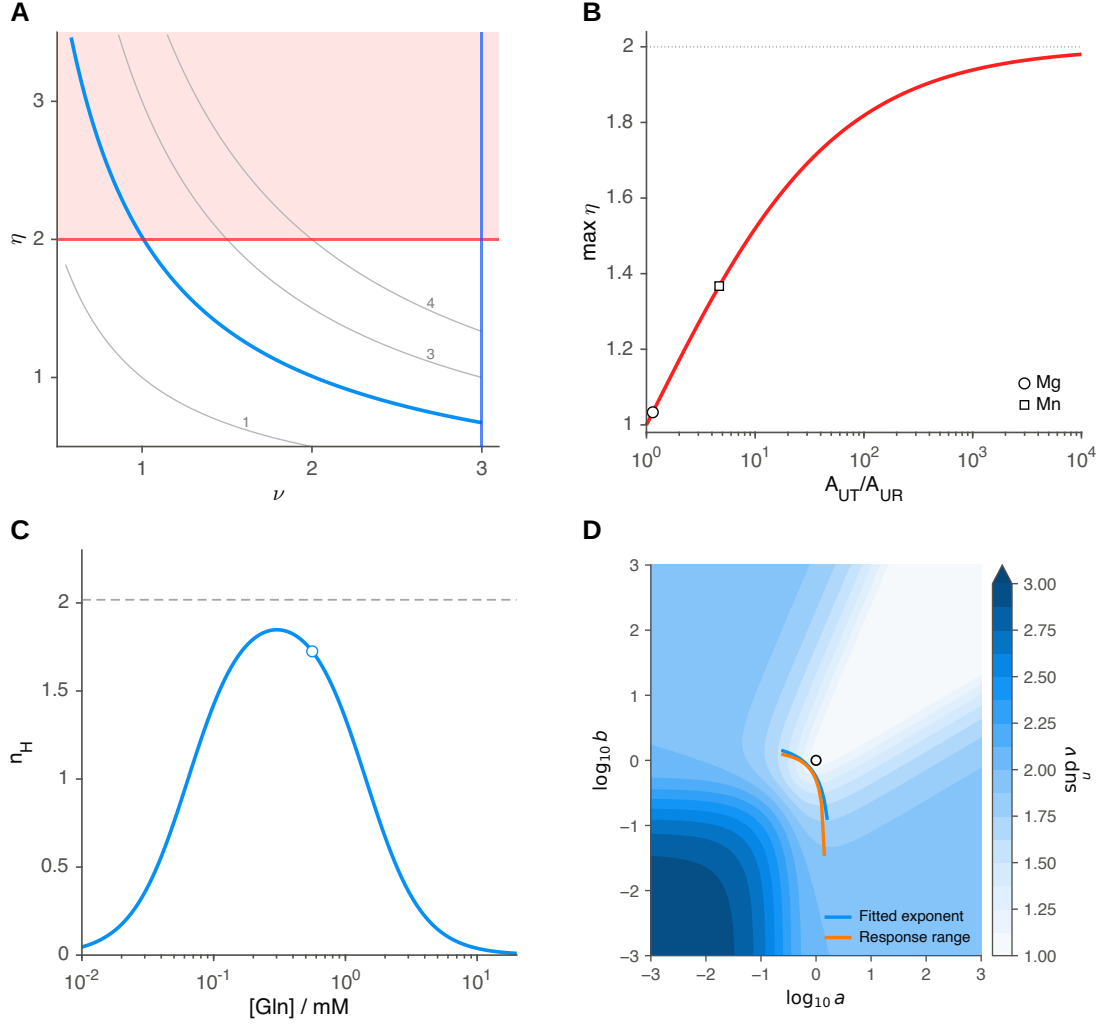


Figure S12: Local factorization and estimator-dependent ladder constraints. (A) Gray contours denote local products $\nu \eta$ of 1, 3, and 4. The blue contour is a numerical reference at 2.018, not a measured local sensitivity. Red marks $\eta=2$, the ceiling of the specified reciprocal switch, and dark blue marks the trimer bound $\nu=3$. (B) Maximum switch elasticity for Eq. (6) when A_{UT}/A_{UR} is at least one. Markers use the Mg and Mn kinetic-constant ratios discussed in the text. (C) Physical-occupancy local slope of the fitted first-layer curve, its fitted exponent of 2.018 (dashed), and midpoint local slope of 1.725 (circle). (D) Three-site ladder coefficients (1,3a,3b,1): distinct parameter loci reproduce coefficient ratio 1.11055 under fitted-exponent (blue) and response-range (orange) estimators. Color gives the maximum local ladder factor over input; the circle is the independent ladder. The loci are conditional calculations and show that the coefficient alone does not identify a unique ladder.